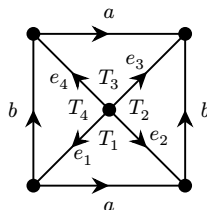


Topology PhD Exam, May 2023

Work the following problems and show all your work. Support all statements to the best of your ability.

1. Let X be a set. A filter base at $x \in X$ is a nonempty collection $\mathcal{U}_x$ of subsets of X containing x such that whenever $U, V \in \mathcal{U}_x$ there exists $W \in \mathcal{U}_x$ with $W \subset U \cap V$. Assume that X has a filter base at each $x \in X$ and that if $U \in \mathcal{U}_x$ and $y \in U$ then $U \in \mathcal{U}_y$. Prove that $\mathcal{B} = \bigcup_{x \in X} \mathcal{U}_x$ is a base for a topology.
2. Consider the torus $T = S^1 \times S^1$ with the following Δ -complex structure.



- (a) Define a 2-chain that you can show is a cycle and which generates $H_2(T)$. The resulting homology class is called a fundamental class of T .
 - (b) Define 1-cochains that you can show are cocycles and which generate $H^1(T)$.
 - (c) Compute a cup product which produces a cohomology class (Kronecker) dual to a fundamental class.
3. Show that the 2-sphere S^2 is not a retract of the real projective plane $\mathbb{RP}^2$, as well as that $\mathbb{RP}^2$ is not a retract of S^2 .
 4. Assume $n > m$. Show that there are no maps from $\mathbb{CP}^n$ to $\mathbb{CP}^m$ that induces a nontrivial map $H^2(\mathbb{CP}^m) \rightarrow H^2(\mathbb{CP}^n)$.
 5. A map $i : A \rightarrow X$ has the homotopy extension property (HEP) for the space Y if for each homotopy $h : A \times I \rightarrow Y$ and each map $f : X \rightarrow Y$ with $f(i(a)) = h(a, 0)$ there exists a homotopy $H : X \times I \rightarrow Y$ with $H(x, 0) = f(x)$ for all $x \in X$ and $H(i(a), t) = h(a, t)$ for all $a \in A$ and all $t \in I$. We call H an extension of h with initial condition f . The map $i : A \rightarrow X$ is a cofibration if it has the HEP for all spaces.

The mapping cylinder of i , denoted $Z(i)$, is the pushout of $i : A \rightarrow X$ and the inclusion $A \rightarrow A \times I$ given by $a \mapsto (a, 0)$.

Prove that the following three statements about i are equivalent:

- (a) i is a cofibration.
- (b) i has the HEP for the mapping cylinder $Z(i)$.
- (c) $Z(i)$ is a retract of $X \times I$.

Answer the following with complete definitions or statements or short proofs.

6. Give the definition of a pushout and prove that if it exists then it is unique up to isomorphism.

7. The Klein bottle, K , may be obtained by taking two Möbius bands, M , and identifying their boundary, $K = M \cup_{\partial M} M$. Apply the Seifert–van Kampen theorem to give a presentation of the fundamental group of K .
8. Describe all of the two-fold coverings of $S^1 \vee S^1$.
9. Prove that for a finite CW-complex X , $H^1(X; \mathbb{Z})$ is torsion-free.
10. Compute the Euler characteristic of the manifold $M = S^4 \times \mathbb{RP}^2 \times \mathbb{CP}^3$.
11. State the Baire Category Theorem.
12. Prove that for $n \neq m$, $\mathbb{R}^n$ is not homeomorphic to $\mathbb{R}^m$.
13. Give an example of a space that is connected but not path connected.
14. State the Urysohn Lemma.
15. Is $\mathbb{R}^\omega$ connected in the uniform topology?